Application of Machine Learning Methods to Evaluate Collectibles

UE24CS352A — Machine Learning Mini-Project | Team 30 | Project ID 50

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1. Problem Statement

Collectible items can have substantially different secondary-market prices. This project investigates whether available product metadata can classify Funko Pop! collectibles into five relative price tiers. The work follows the problem framing in Xiaotian Lu's Stanford CS229 report, "Application of Machine Learning Methods to Evaluate Collectibles" (2021).

2. Dataset

The local cleaned dataset contains 359 unique products from the curated CSV in the original author's public CS229 repository. The dataset categories are Anime (198), Marvel (69), Ad Icons (44), Disney (37), and Game of Thrones (11). The target source column is average_sale_price. In the local run, prices had a mean of 85.98, median of 47.00, standard deviation of 168.82, minimum of 14.00, and maximum of 2,150.00 in the dataset's reported units.

The Stanford paper reports 358 products after filtering, while this run has 359. This discrepancy is documented rather than treated as an exact reproduction. The precise definition of the price field and the source repository's data-use terms should be verified before redistribution.

Source repository: https://github.com/timothyxlu/CS229/

Curated CSV: https://raw.githubusercontent.com/timothyxlu/CS229/master/project/src/data_sourcing/data_curated.csv

3. Methodology

Records missing product names, categories, or prices and records with non-positive prices were removed. Repeated product names were aggregated using median price. Product-name indicators were derived for terms such as metallic, glow, flocked, chase, exclusive, SDCC, limited, glitter, and chrome. Available metadata included category, material, special version, release information, and search volume where present.

Five relative price tiers were created using quantiles computed from the training split, with approximate thresholds of 26.42, 35.84, 53.96, and 86.97. Six configurations were compared: a majority-class baseline, Linear SVM, RBF SVM, Decision Tree, Bagged Trees, and Random Forest. Evaluation used a held-out test set of 90 products. Metrics were accuracy, balanced accuracy, macro-F1, and weighted-F1; macro-F1 was used to select the leading model because it reflects performance across all classes.

4. Results

Model	Accuracy	Balanced accuracy	Macro-F1	Weighted-F1
RBF SVM	0.3778	0.3481	0.3448	0.3742
Random Forest	0.3667	0.3493	0.3425	0.3660
Linear SVM	0.3667	0.3282	0.3282	0.3693
Bagged Trees	0.3222	0.3128	0.3082	0.3224
Decision Tree	0.2889	0.3035	0.2884	0.2872
Majority baseline	0.2333	0.2000	0.0757	0.0883

RBF SVM achieved the highest macro-F1 (0.3448) and 37.78% test accuracy. Random Forest had marginally higher balanced accuracy (0.3493 versus 0.3481). The RBF SVM correctly classified 34 of 90 test products. Its recall was strongest for Tier 5 (0.609) and Tier 1 (0.476), and weakest for Tier 4 (0.211) and Tier 3 (0.214). The results suggest that the middle price tiers are difficult to distinguish.

5. Demonstration

The command-line predictor classified “Funko Pop Animation Dragon Ball Z Beerus Metallic SDCC Figure 120” as Tier 5/5. This is a relative category prediction, not an exact price estimate or an independently verified market valuation.

6. Conclusion and Limitations

SVM and tree-based ensembles outperformed the majority baseline on this single holdout split, but overall predictive performance was modest. The small dataset, category imbalance, overlap between price tiers, and reliance on a single test split limit generalization. Price tiers are relative to this dataset and are not universal market bands. Further work should verify data provenance and usage terms, investigate the 358-versus-359 product discrepancy, audit the feature pipeline for leakage, and evaluate with repeated stratified cross-validation and additional real observations.

Reference

Lu, X. (2021). *Application of Machine Learning Methods to Evaluate Collectibles*. Stanford CS229 project report. https://cs229.stanford.edu/proj2021spr/report2/81980743.pdf